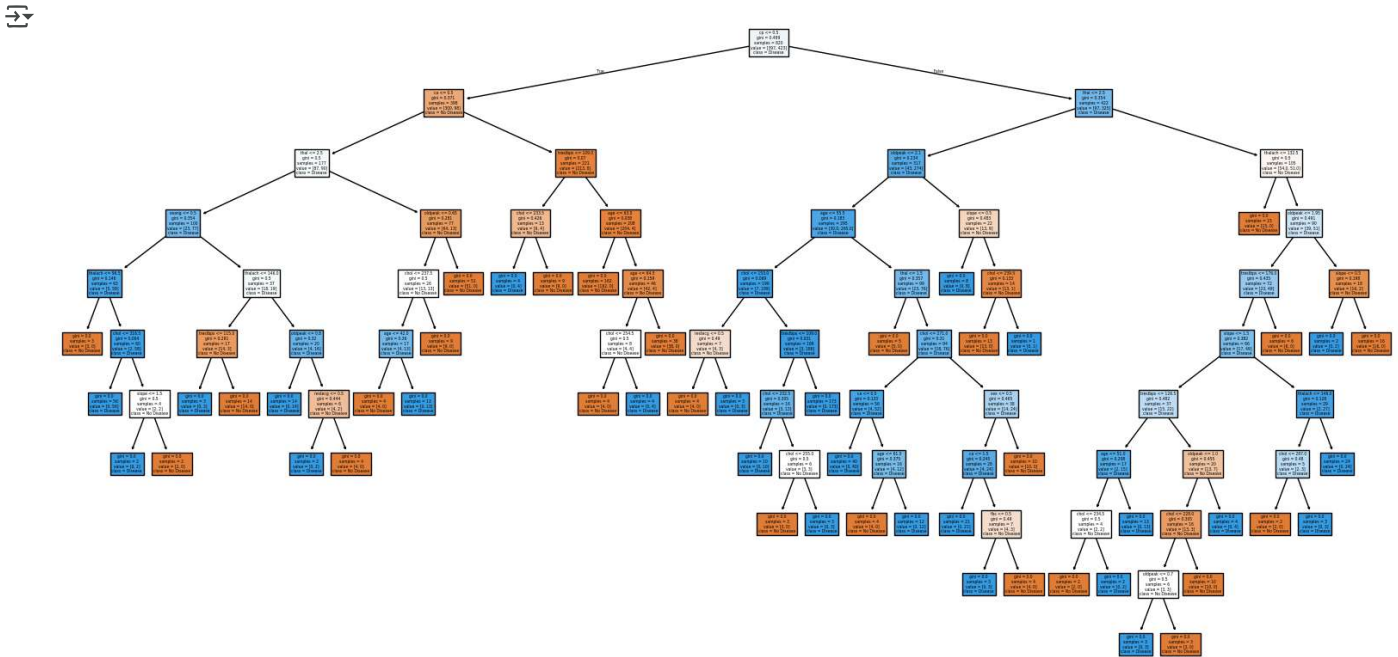


```

1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3
4 df = pd.read_csv("/content/heart.csv")
5 X = df.drop("target", axis=1)
6 y = df["target"]
7
8 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
9
10 from sklearn.tree import DecisionTreeClassifier, plot_tree
11 import matplotlib.pyplot as plt
12
13 dt = DecisionTreeClassifier(random_state=42)
14 dt.fit(X_train, y_train)
15
16 plt.figure(figsize=(20,10))
17 plot_tree(dt, filled=True, feature_names=X.columns, class_names=["No Disease", "Disease"])
18 plt.show()
19

```



```

1 print("Train Accuracy:", dt.score(X_train, y_train))
2 print("Test Accuracy:", dt.score(X_test, y_test))
3
4 dt_limited = DecisionTreeClassifier(max_depth=4, random_state=42)
5 dt_limited.fit(X_train, y_train)
6
7 print("Train Accuracy (limited):", dt_limited.score(X_train, y_train))
8 print("Test Accuracy (limited):", dt_limited.score(X_test, y_test))
9

```

```

Train Accuracy: 1.0
Test Accuracy: 0.9853658536585366
Train Accuracy (limited): 0.8829268292682927
Test Accuracy (limited): 0.8

```

```

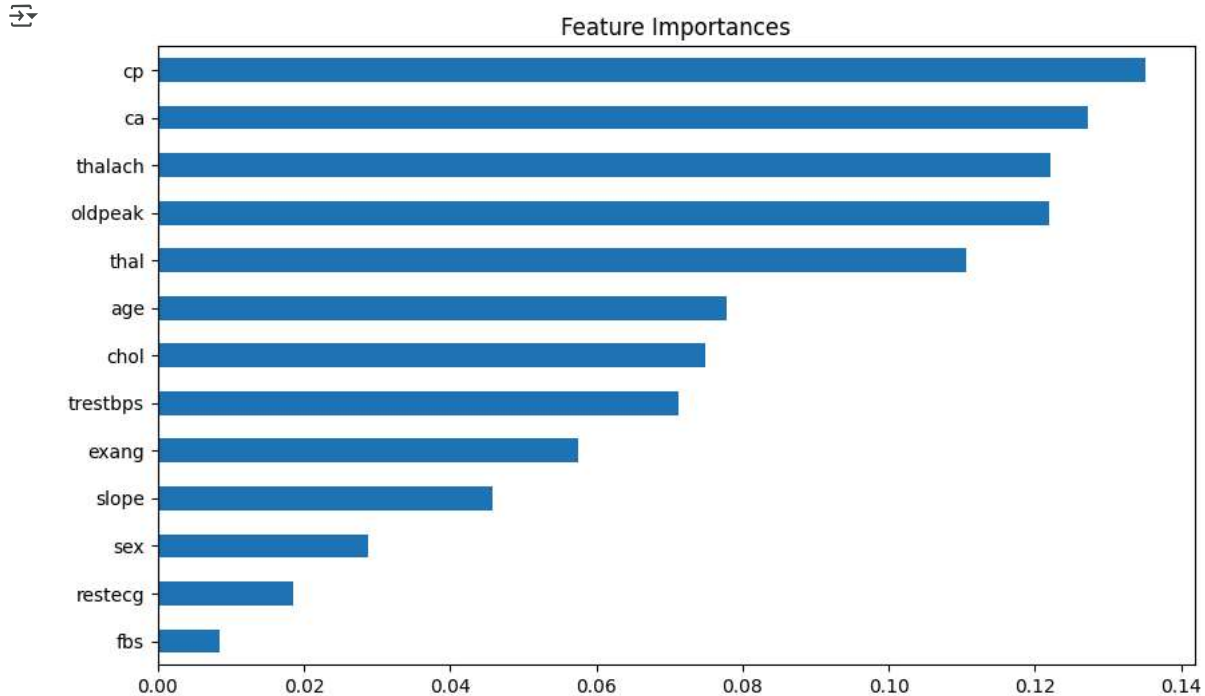
1 from sklearn.ensemble import RandomForestClassifier
2
3 rf = RandomForestClassifier(n_estimators=100, random_state=42)
4 rf.fit(X_train, y_train)
5
6 print("Random Forest Train Accuracy:", rf.score(X_train, y_train))

```

```
7 print("Random Forest Test Accuracy:", rf.score(X_test, y_test))
8
```

↗ Random Forest Train Accuracy: 1.0
Random Forest Test Accuracy: 0.9853658536585366

```
1 import seaborn as sns
2
3 importances = pd.Series(rf.feature_importances_, index=X.columns)
4 importances.sort_values().plot(kind='barh', figsize=(10,6), title="Feature Importances")
5 plt.show()
6
```



```
1 from sklearn.model_selection import cross_val_score
2
3 scores = cross_val_score(rf, X, y, cv=5)
4 print("Cross-Validation Accuracy:", scores.mean())
5
```

↗ Cross-Validation Accuracy: 0.9970731707317073